Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics

2.2 – Newton's 2nd Law

Unit 2: Forces and Motion

Before you watch

A skydiver jumps from a plane. Her speed changes several times before she lands — speeding up, then steady, then slowing down, then steady again. This video is only about three and a half minutes long. Watch it once all the way through, then we go back over it together.

Listen for one pair of words: **balanced** and **unbalanced**.

While you watch

1. Name the **two** forces acting on the skydiver as she falls.

   _____

2. One of those two forces stays the same the whole time. Which one?

   _____

3. The other force changes as she falls. Does it get bigger or smaller as her speed increases?

   _____

Newton's Second Law

> **When the forces on an object are unbalanced, the object accelerates — it speeds up, slows down, or changes direction — in the direction of the *bigger* force. When the forces are balanced, there is no acceleration: the object's speed stays constant.**

Check: "balanced" does not mean "stopped." A skydiver falling at a steady speed has balanced forces *and* is still moving — Newton's First Law again. Newton's Second Law is about what happens the moment the forces *stop* being balanced.

The skydive, in four stages

Fill in the last column yourself, using **speeding up**, **constant speed**, or **slowing down**.

Stage	Forces	What happens to her speed
Just after jumping	gravity much bigger than air resistance	
Falling for a while	air resistance has grown until it equals gravity	
Parachute opens	air resistance suddenly much bigger than gravity	
After the parachute has been open a while	air resistance has shrunk back down until it equals gravity again	

Apply it

1) **State** Newton's Second Law, in your own words.

State: Give a specific name, value or other brief answer without explanation or calculation.

2) **Describe** how the two forces on the skydiver compare in the first few seconds after she jumps, and what happens to her speed because of it.

Describe: Give a detailed account or picture of a situation, event, pattern or process.

3) **Explain** why the skydiver's falling speed stops increasing even though gravity has not changed.

Explain: Give a detailed account including reasons or causes.

Extension

Think further. When the parachute first opens, the skydiver is briefly pushed *upward* in her seat straps, even though she is still moving downward overall. Using Newton's Second Law, suggest why.